R Markdown

Introduction

R Markdown is built into RStudio and allows you to create documents (in HTML, PDF, and Word formats), presentations (HTML or PDF formats) or webpages from R. With R Markdown, you can embed R code into your documents.

Why use R Markdown?

- Turn work in R into more accessible formats
- Incorporate R code and R plots into documents
- R Markdown documents are reproducible – the source code gets rerun every time a document is generated, so if data change or source code changes, the output in the document will change with it.

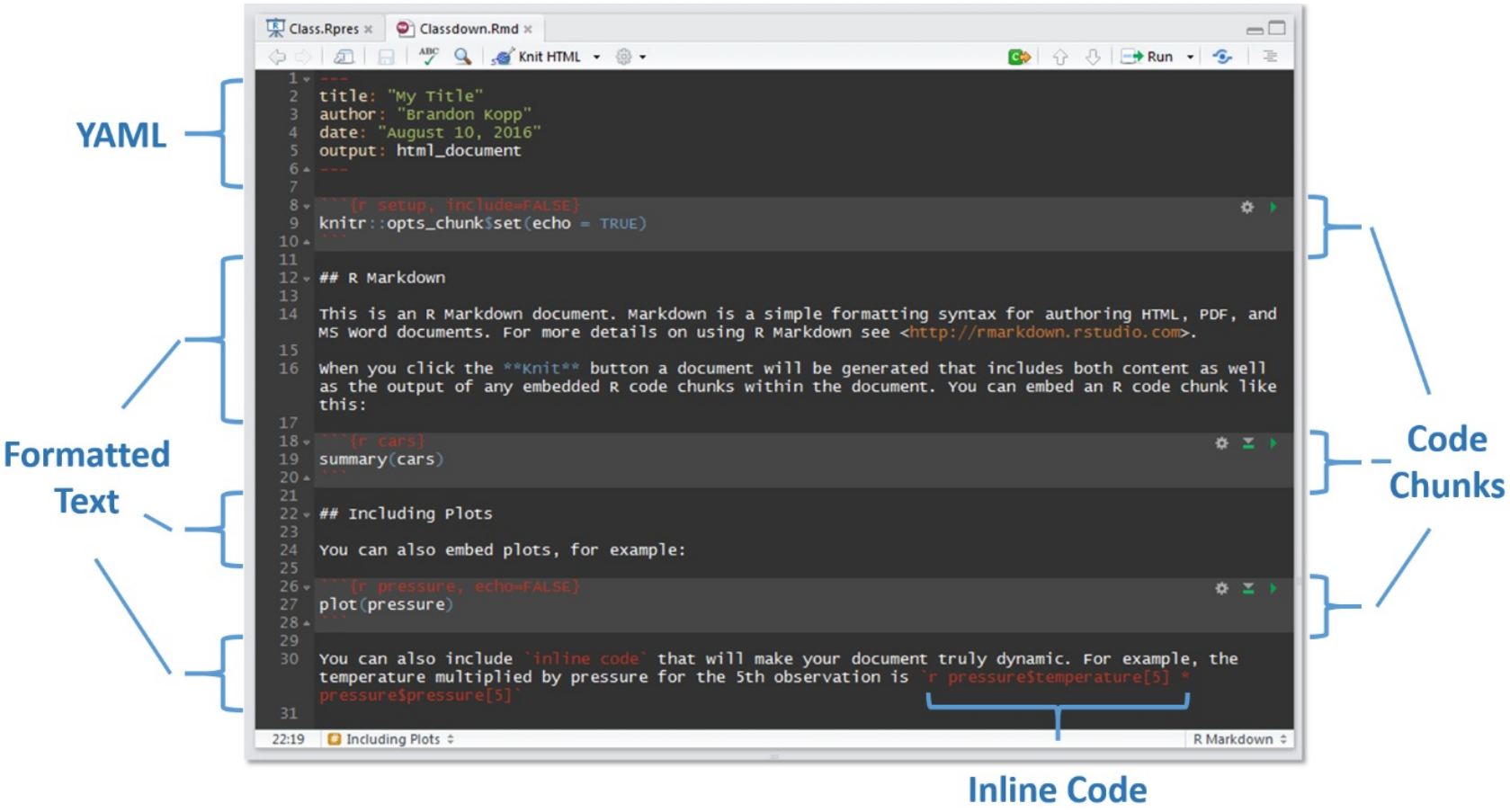
Opening an R Markdown Document

1. Open RStudio
2. File > New File > R Markdown
3. Enter Title, Author, and Output Format (can be changed later)
4. Click OK
5. Comes with an example that can be altered or deleted

Note: While it may not be necessary for newer versions of RStudio, it is a good idea to install:

- knitr & rmarkdown packages
- some distribution of the LaTeX system (e.g., MiKTeX)
  - LaTeX is a typesetting system that is needed to convert R Markdown into formatted text for PDFs

RMarkdown Sections



YAML

```yaml
---
title: "Title"
author: "Kalyan R"
date: "Nov 3, 2019"
output: pdf_document
fontsize: 11pt
urlcolor: "red"
---
```

- Stands for "YAML Ain't Markup Language"
- YAML header is where you set options to fine tune the overall output of your document
- Set of key:value pairs that describes the file that should be built from the R Markdown file
- you can set: font (size and style), default figure options (height, width, etc.), theme, etc
- YAML header options for PDF

Text Syntax

R Markdown offers shorthand for formatting text.

- *italics* or _italics_ —> *italics*
- **bold** or __bold__ —> **bold**
- [BLS](www.bls.gov) —> BLS
- superscript^2^ —> superscript^2
- subscript~2~ —> subscript$_2$
- ~~strikethrough~~ —> ~~strikethrough~~
- image:
- inline math symbols: π -> π

Lot more options available. Check the links in references

List - Unordered List

Syntax:

```
- Item
+ Sub-item
- Item
- Item
```

Output:

- Item
  - Sub-Item
- Item
- Item

List - Ordered List

Syntax:

```
1. Item Number 1
2. Item Number 2
+ Sub-item
3. Item Number 3
```

Output:

1. Item Number 1
2. Item Number 2
   - Sub-item
3. Item Number 3

Line Breaks

- Line breaks in markdown can be confusing
- In order to to create a line break, end a line with two spaces
- You could also use the LaTeX command \newline

Code Syntax

- Synatx for R code chunks is:

  ```
  ```{r}
  ```
  ```

- **Example:** The following R code

  ```
  ```{r}
  paste("Hello", "World!")
  ```
  ```

  produces the following output

```
## [1] "Hello World!"
```

- Chunk options:
  - `echo=FALSE` prevents source code from being displayed
  - `eval=FALSE` prevents source code from being executed
  - **Example:** ```{r, eval=TRUE, echo=FALSE}
  - Other chunk options: warning, message, result, etc

- Inline R code: ` r paste("Hello", "World!") `

Code chunks to make tables (1)

```
mycars <- mtcars[1:2,1:6]
mycars
```

```
##                mpg cyl disp  hp drat    wt
## Mazda RX4       21   6  160 110  3.9 2.620
## Mazda RX4 Wag   21   6  160 110  3.9 2.875
```

Code chunks to make tables (2)

We can use xtable packages to make better table

```
options(xtable.comment = FALSE)
out <- xtable::xtable(mycars,caption = "Motor Trend Car Tab
print(out, type='latex', html.table.attributes="border=2")
```

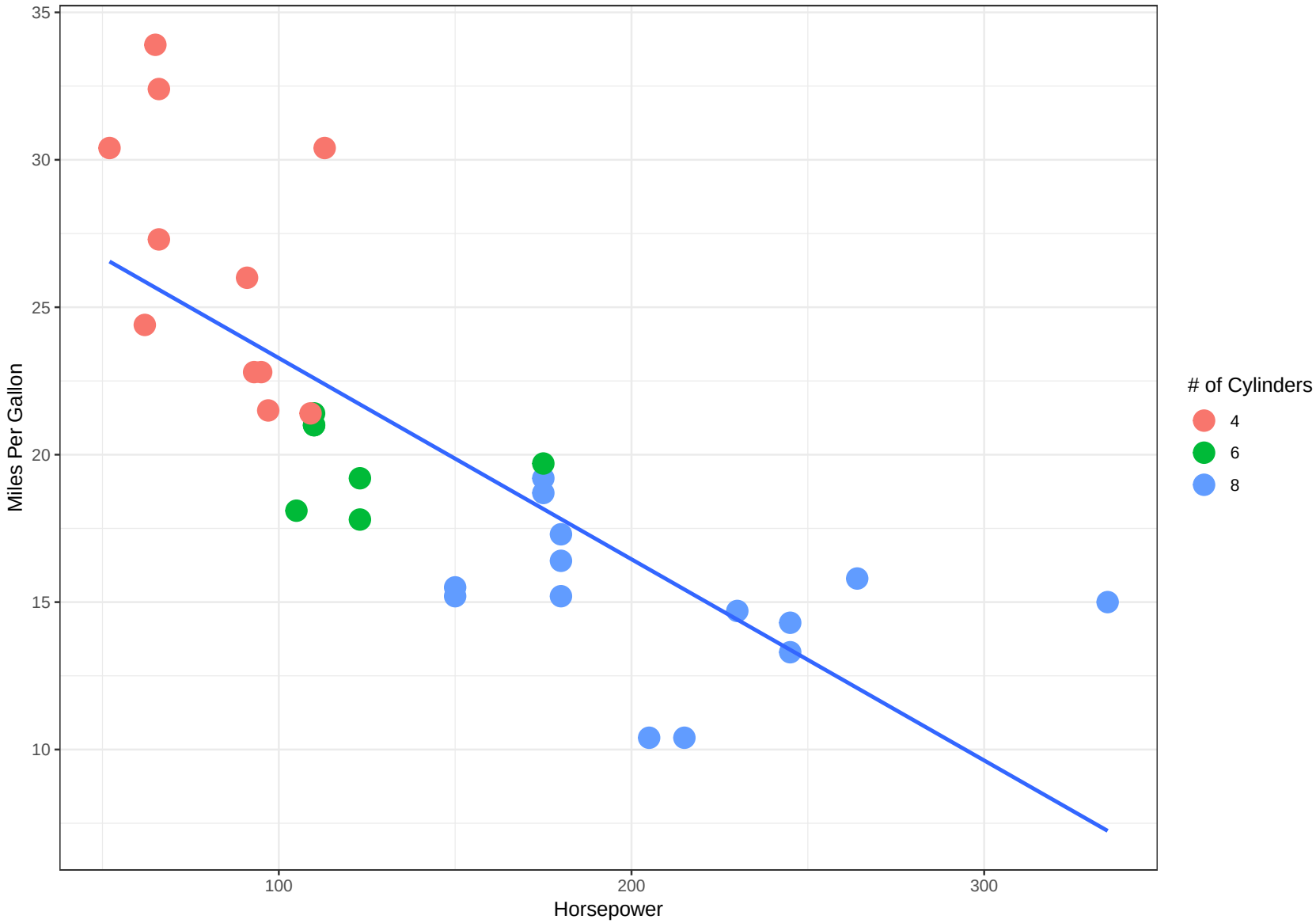
	mpg	cyl	disp	hp	drat	wt
Mazda RX4	21.00	6.00	160.00	110.00	3.90	2.62
Mazda RX4 Wag	21.00	6.00	160.00	110.00	3.90	2.88

Table 1: Motor Trend Car Table

Code chunks to make figures (1)

```
library(ggplot2)
gg <- ggplot(mtcars, aes(hp, mpg)) +
        geom_point(aes(color=as.factor(cyl)), size=5) +
        geom_smooth(method="lm", se=FALSE) +
        labs(x = "Horsepower",y= "Miles Per Gallon",
             color= "# of Cylinders") +
        theme_bw()
gg
```

Code chunks to make figures (2)



Useful References

- R Markdown: The Definitive Guide (Book)
- R Markdown Reference Guide (PDF)
- R Markdown by RStudio